MS2 Summary — AuroraCast

1. What I achieved in this milestone

What works now that did not before

- Live NOAA RTSW solar wind is stored hourly from all spacecraft, with data from 14 Oct. It runs on the Infomaniak VM, with GitHub Actions into Tiger Cloud as an independent backup.
- GFZ Kp is polled hourly, and every change is recorded with the time I saw it (`kp_revisions`), since 31 Oct.
- History is in the store on the VM: OMNI 1-min 1998 to 15 Oct 2026 (12.8 M rows before 2024) and Kp from 1998.
- The EDA is done on 1998-2023 only: both persistence variants, storm base rate, OMNI coverage, first signal check.
- Targets and decision thresholds are frozen in the README, before anything was scored on the test years.

Key decisions, and why (full reasoning in `docs/DECISIONS.md`)

- Cron and a Python loop instead of Airflow for now: the reviews named the stack as the main risk, and the pipelines are plain Python, so Airflow can still wrap them later.
- Tiger Cloud came first because the poller could not wait for a VM and my laptop is not always on. The server deal fell through, so I used the free-credit Infomaniak VM.
- I keep rows from all spacecraft and filter later, because live data cannot be fetched again.
- I replaced a planned "first value seen" column with a revision table after seeing that GFZ publishes preliminary values while an interval is still running.

What to look at first

1. README: "Frozen targets and decision thresholds", then "What runs where".
2. `notebooks/01_eda.ipynb`: the findings section near the end.
3. `src/auroracast/ingest/` (`rtsw.py`, `kp.py`, `omni.py`) and `.github/workflows/rtsw-poller.yml`.

What this cost me

- I was sick from early to mid October, so the poller started on 14 Oct instead of early October. RTSW from before then is lost for good.
- Kp publication times before 31 Oct were never recorded and cannot be reconstructed.
- `features.py` and the feature table are not built, so no model exists yet. This is the first job for MS3.
- The frozen baseline is hard to beat. It ignores the publication delay, and solar wind barely tracks the change in Kp (|rho| at most 0.11, training years). I may miss the 10 % target.

2. Response to MS1 reviews

A1, B3 · Persistence unmeasured, moving goalposts — Changed (`dc21a87`). Measured both ways on 1998-2023: last completed MAE 0.627, last published 0.806; storm base rate 2.64 %. Targets are pinned to the stronger variant and frozen in the README with the decision threshold. The proposal's storm target was weaker than persistence (0.52 / 0.52), so it now only acts as a floor.

A2, B4 · Give NOAA's forecast a target; originality — Partly (README, "Frozen rules"). Live storm recall at equal precision against NOAA is now part of the success criteria. Logging NOAA's forecast and showing it on the dashboard are not built yet. On originality: I agree with the editors that NOAA publishing a forecast is not a problem. It is the bar to beat, and my contribution is the automated pipeline with a measured comparison against it.

A3, B1 · Train/serve skew not quantified — Partly (`notebooks/01_eda.ipynb`, section 6). The measurement strategy has three parts:

- RTSW vs OMNI on the same minutes, repeated whenever NASA extends its 2026 file.
- Nowcast vs definitive Kp from `kp_revisions`, which records every value I see.
- Offline test MAE vs the live 30-day MAE, once forecasts run.

First numbers from only 33 overlapping hours, hourly means: speed 16.6 km/s (r 0.93), density 1.17 /cc (r 0.79), Bz 1.21 nT (r 0.64). Too short to conclude, and OMNI uses Wind while RTSW uses SOLAR1. Nowcast vs definitive has no data yet, because no interval has both values.

A4 · Self-hosted stack too heavy, provider open — Changed (`1473a8e`, `docs/DECISIONS.md`). Provider decided: Infomaniak. Airflow and an MLflow server moved to stretch, so the planned core is TimescaleDB, a cron scheduler and Streamlit. Today only TimescaleDB runs, with a loop and a crontab entry for polling.

B2 · Few independent storms in the test years — Partly (README; `notebooks/01_eda.ipynb`, section 7, `src/auroracast/baseline.py`). Stated: 61 independent events in 2024-2025 (storm intervals under 24h apart count as one). Only the count was taken from those years. Because recall over so few events is noisy, the final evaluation will report the per-event hit rate out of 61, with a confidence interval from resampling whole storm events.

E1 · Start the poller now — Changed (`72a26af`, `a850d57`). Hourly polling with persistent storage from the first run, on GitHub Actions while no VM existed, as you suggested. It started later than you asked, as noted above.

E2 · Test the publication-delay rule — Not yet. It will be one tested function in `features.py` for MS3. The delay data it needs is being collected: so far 23 % of 39 intervals were first seen before the interval ended, median 4.7h after its start (upper limits, because runs are irregular). `src/auroracast/baseline.py` only defines the two lags for the EDA.

E3 · Proposal file — Changed (`1b8b78d`, `docs/proposal.pdf`). Re-exported with a text layer and renamed as requested.